

The background features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, creating a modern, layered effect. The shapes are concentrated on the left and right sides of the frame, leaving a central white area.

rEdOx

The Problem

- ▶ Predict ligand placement and H12, especially across agonist and antagonist states
- ▶ 57 structures labelled agonist and 21 antagonist
- ▶ **19 of 21 antagonists carry C455H**
- ▶ 77 usable structures, 16 selected for validation (4 antagonist)

The Ideas

Binary classification
using a chemistry
derived agonist and
antagonist state prior
to steer the model

Validation set on
benchmark to select
favorable structures
to generate synthetic
data

Selective structural
finetuning

The Plan

Binary classification
using a chemistry
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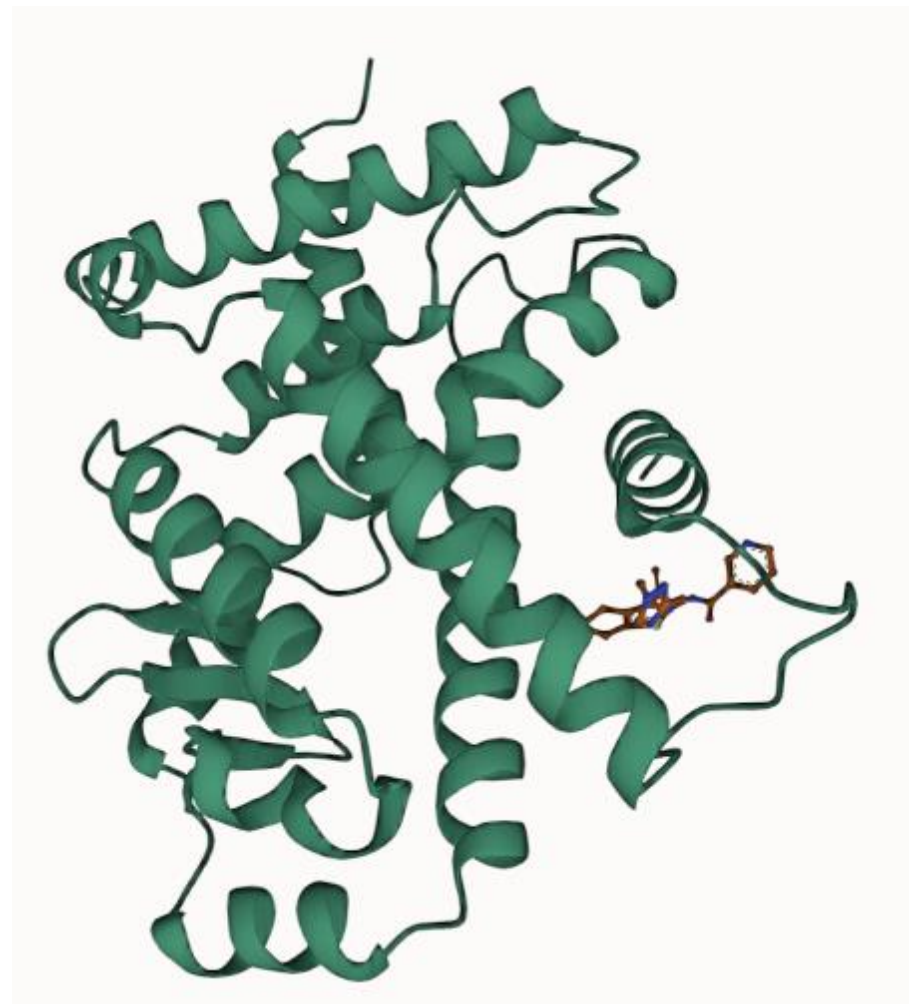
Selective structural
finetuning

MSA Baseline

- ▶ OpenFold needs MSAs - Foundry can run without, but the results aren't good
 - ▶ Agonist 5NI5_A, H12 RMSD 17.93A to 0.37A
- ▶ 16 val split used with baseline model, with MSA
 - ▶ GDT-HA 0.927, 0.89 on antagonists
 - ▶ Mean interface 0.913, ligand bisyRMSD 0.904A

Synthetic Structures

- ▶ ChEMBL ligands deduplicated on scaffold and filtered on novelty to PDB ligands
- ▶ Sampled all 27 unique sequences in the PDB data
- ▶ Generated 10 diffusion samples for 10 seeds
- ▶ Used best sample per model, $N = 400$

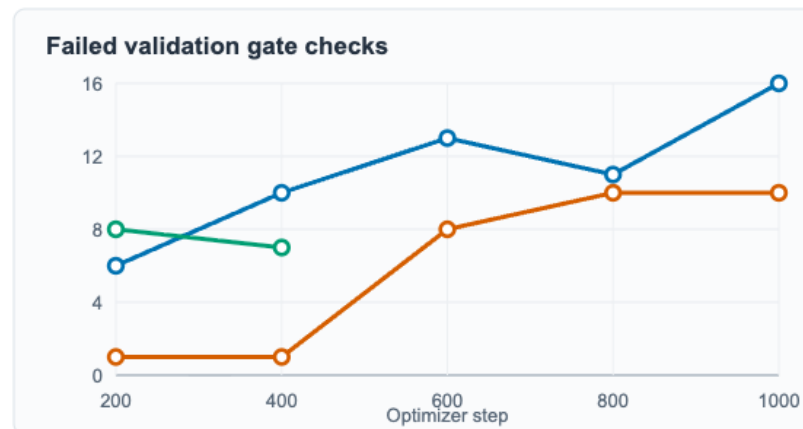
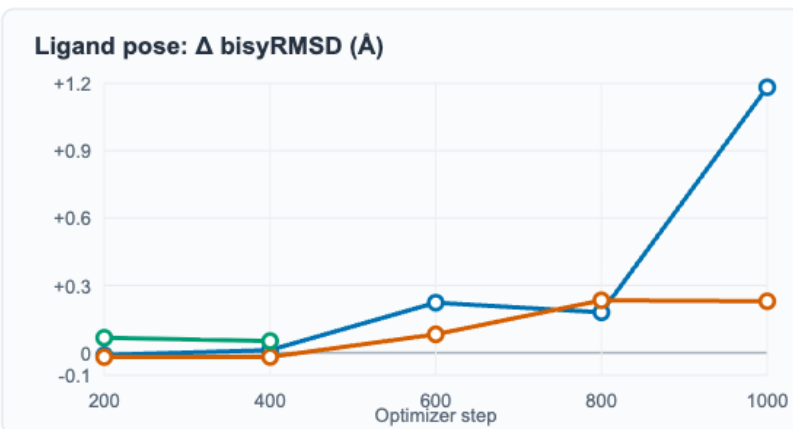
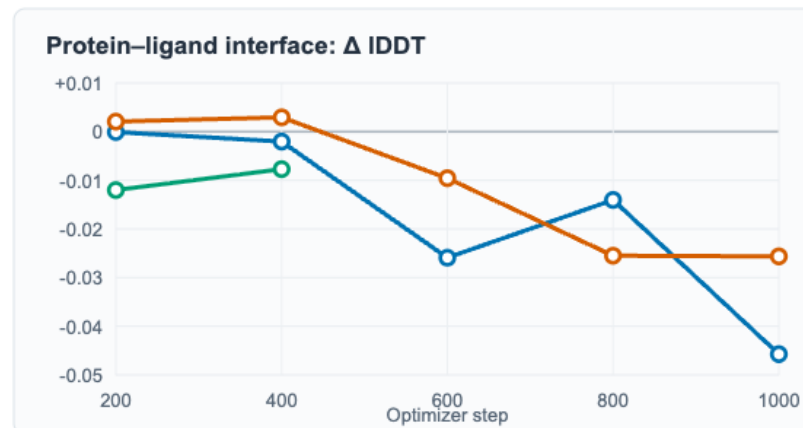
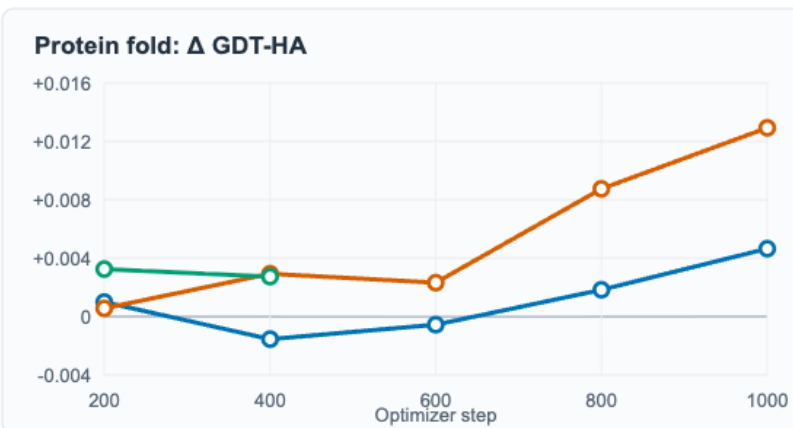


Fine-tuning scored against 16 val complexes

—●— **Balanced rehearsal**
LR 1×10^{-4} · crop 192

—●— **PDB-heavy rehearsal**
LR 3×10^{-5} · crop 256

—●— **Agonist-heavy rehearsal**
LR 3×10^{-5} · crop 256



Higher GDT-HA and interface IDDT are better; lower bisyRMSD and fewer failed checks are better.

All plotted checkpoints retained 16/16 correct H12 state calls. PDB-heavy step 200 also had one train gate failure (46 complexes).

Learning rate changes alongside PDB rehearsal mix and crop size; curves do not isolate a learning-rate effect. No test scores are used.

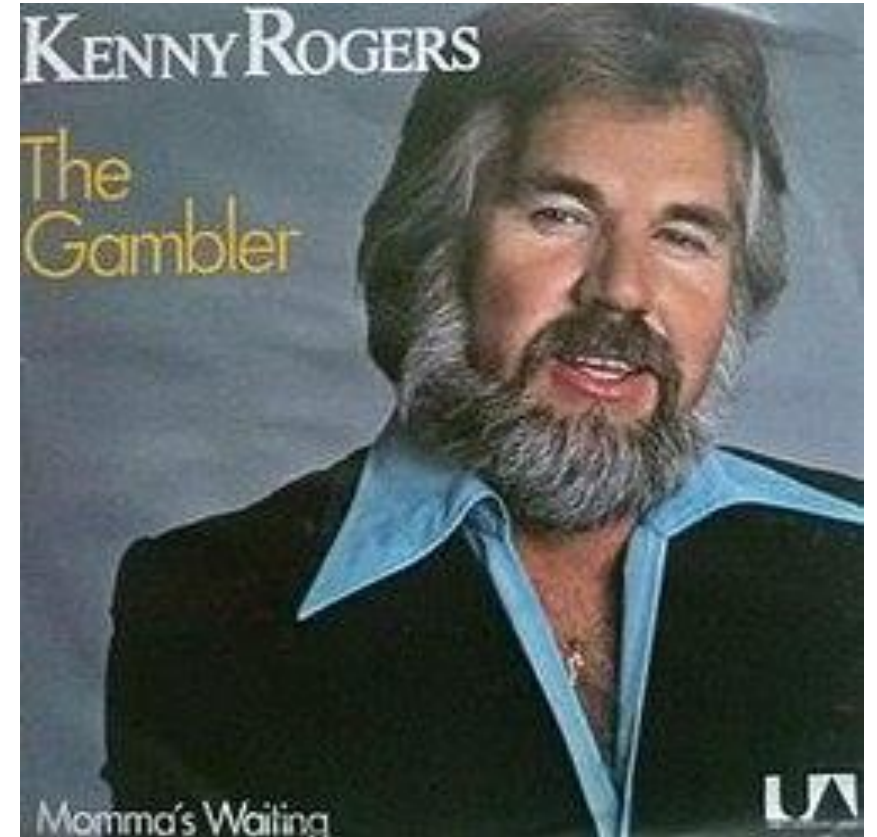
Conducting a CLAUDEtopsy

- ▶ **Foundry Interface:** The installed finetune-openfold3 image *does* support the more config fields through direct module-run, but the supplied Nextflow entrypoint does not expose them.
 - ▶ **Version Mismatch:** dev-catalog:apheris-openfold3-v0.15.0 != v0.15.1 image installed
 - ▶ Kit's H12-only reference call reverses two deposited antagonists (76/78)
 - ▶ `compute-metrics` intermittent late-write crash under concurrent load
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- ▶ Saying anything in ~/data can be used, and having it as a read-only directory before agentic exploration is a good idea, if you can ensure that no one deposits the real data in there without you realising!!!

Extra Slides

Key Takeaways

- Know when to hold 'em, know when to fold 'em



Agentic Crashes

- ▶ No, not that kind. More like this ->
- ▶ Using the same Codex/Claude in the same local directories is tricky!
- ▶ Pro: can access your partners work
- ▶ Con: Collisions
- ▶ Careful considerations required

